

Ligjerata 5d

Momenti i inercisë

$$\Delta m$$

$$\vec{M} = \vec{M}_1 + \vec{M}_2 + \vec{M}_3 + \dots + \vec{M}_n = \sum_{i=1}^n \vec{M}_i$$

$$\vec{M} = \sum_{i=1}^n \vec{F}_i \cdot \vec{r}_i$$

$$\vec{F}_i = \Delta m_i \vec{a}_i$$

$$\vec{M} = \sum_{i=1}^n \Delta m_i \vec{a}_i \vec{r}_i$$

$a_i = r_i \alpha$
 $\vec{a}_i = \vec{r}_i \vec{\alpha}$

ngjithmshëndor

$$\vec{M} = \sum_{i=1}^n \Delta m_i \vec{\alpha} \cdot r_i^2$$

$$\vec{M} = \vec{\alpha} \underbrace{\sum_{i=1}^n m_i r_i^2}_{\text{momenti i masës te inercisë}}$$

$$I = \sum_{i=1}^n m_i r_i^2$$

$$\vec{M} = \langle I \rangle \vec{\alpha} \quad \vec{\Gamma} = \omega \vec{a}$$

$$\Delta m \rightarrow 0$$

$$\lim_{\Delta m \rightarrow 0} \langle I \rangle = \lim_{\Delta m \rightarrow 0} \sum_{i=1}^n r_i^2 \Delta m$$

$$dI = r^2 dm / \int$$

$$\int dI = r^2 \int dm$$

$$I = r^2 \int dm$$

$$\rho = \frac{m}{V}$$

$$\rho = \frac{dm}{dV}$$

$$dm = \rho \cdot dV$$

$$I = \int r^2 \rho dV$$

